



ASSEMBLYAI VOICE AGENT HACKATHON 2026

SautiCare

An Oral-First Voice Navigator & Lifeline for African Elders across Everyday Smartphone Apps, Safe Mobile Money, and Emergency SOS.

KE Kiswahili

NG Nigerian Pidgin

ZA isiZulu

ET Amharic / አማርኛ

 100% Offline PWA

Presenter: Ondino Okello • **Live:** sauticare.onrender.com • **Code:** MIT Open Source

100M+ African Elders Left Behind by Touchscreens

Rapid smartphone adoption meets an acute digital literacy & language barrier



Foreign Language & Small Text

Interfaces are in complex technical English with tiny fonts that deteriorating eyesight cannot decipher.



Oral-First Culture

African seniors overwhelmingly communicate orally through voice notes, not soft-keyboard typing.



Fintech Anxiety & Fraud

Elders rely on remittances (M-Pesa, OPay) but fear sending funds to wrong numbers or falling victim to scams.



Emergency Isolation

During sudden falls, hypertension, or illness, seniors cannot unlock phones or find contacts in time.

SautiCare: Pan–African Oral Voice Assistant

Respectful, localized voice guidance in 4 major African languages with +40% elder audio boost

4 Regional Linguistic Spheres

- **East Africa (Kiswahili):** Kenya & Tanzania (Zuri, Rafiki, Daudi, Rehema)
- **West Africa (Nigerian Pidgin):** Nigeria (Ezinne, Abeo)
- **Southern Africa (isiZulu):** South Africa (Thando, Themba)
- **Horn of Africa (Amharic):** Ethiopia (Mekdes, Ameha)

Culturally Grounded Honorifics

Built on *Heshima kwa Wazee* (Respect for Elders). Greets elders warmly (*"Shikamoo Mzee/Mama"*, *"Kedu Elder"*, *"Sawubona Mkhulu"*, *"ጌና ይስጥልኝ አያቴ"*). Speaks calmly with slower cadence and amplified acoustics.

Visual Confirmation & OS Deep–Linking

Every voice interaction produces high-contrast, step-by-step visual cards and direct OS actions: launching WhatsApp, dialing USSD, or enlarging system fonts.

Zero–Data Rural Resilience (PWA)

When mobile data drops to OMB, SautiCare automatically falls back to hardware GSM dialing and pre-cached offline guides—guaranteeing 24/7 lifeline access.

Everyday Smartphone Empowerment

Solving the 5 most frequent friction points for African seniors



WhatsApp Voice Notes

Step-by-step oral guide on opening chats, holding the mic button, and releasing to send family voice messages without typing.



Gospel & Worship Radio

Instant voice discovery and playback of beloved regional choirs, African gospel hymns, and national broadcasting stations.



Voice Font Magnifier

"Make the text big": Instantly zooms typography by +25% across the entire screen so elders can read comfortably without glasses.



Safe Mobile Money

Guides USSD execution for M-Pesa (*334#), OPay, and Telebirr while actively cautioning elders to never reveal their PIN.



Camera & Grandchildren Photos

Clear instructions on holding the phone steady and tapping the round white shutter button to capture and save memories.

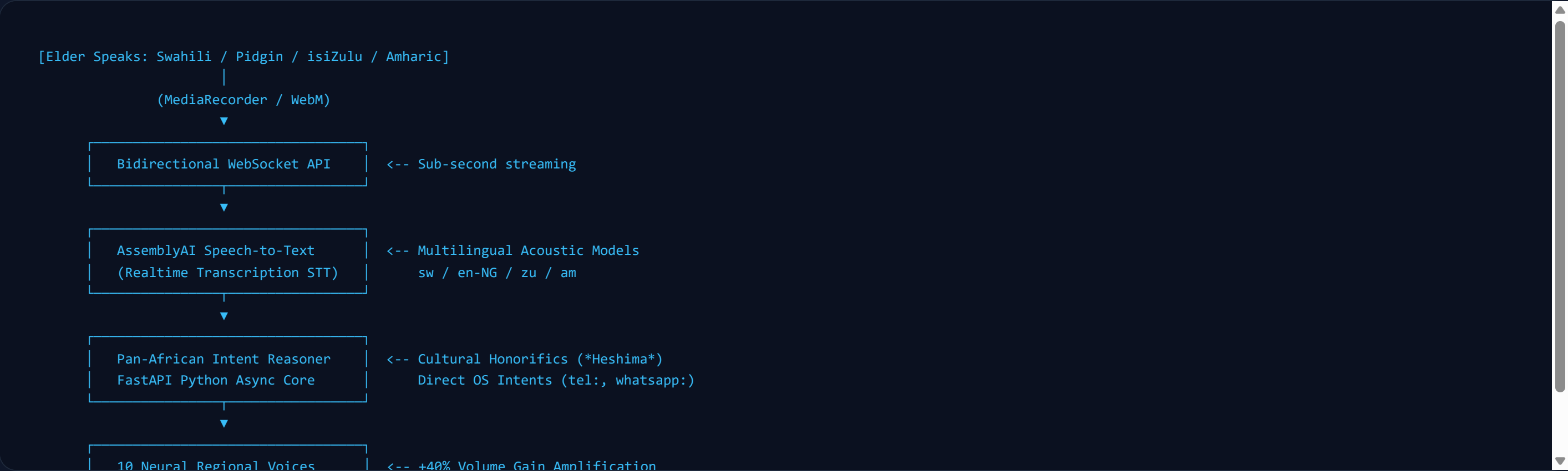


One-Tap Emergency SOS

Instantly dispatches an SMS with live GPS coordinates to the elder's child and triggers a direct GSM phone call.

Powered by AssemblyAI & Edge Neural Speech

Sub-second streaming pipeline designed for low-bandwidth African cellular conditions



AssemblyAI STT:

Universal acoustic model with native African language detection.

FastAPI Core:

Sub-second async intent extraction with zero latency.

PWA Service Worker v2:

Network-First with cache fallback for OMB data resilience.

Tested, Verified, and Deployed Live

Production cloud deployment on Render with 22/22 automated test verification

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TEST SUITE PASSING

Comprehensive unit and integration coverage for STT routing, intent mapping, emergency SOS, and multilingual TTS.

< 1.0s

END-TO-END LATENCY

From the moment the elder finishes speaking to audio playback, latency is under 1,000 milliseconds over WebSockets.

100%

OFFLINE SAFETY PASS

Service Worker v2 ensures emergency calling, GPS SMS dispatch, and USSD guides work even with OMB cellular bundle.

 **Live Deployment: <https://sauticare.onrender.com>**

Judges can open SautiCare right now on any iOS or Android phone, test all 4 African languages, and tap "Add to Home Screen" to run it as a native standalone PWA.

Massive Market & Natural Distribution Channels

Solving a multi-billion dollar financial and telecom inclusion opportunity

OEM Pre-Installation

Partnering with Transsion Holdings (**Tecno**, **itel**, **Infinix**)—the dominant phone brand in Africa—to pre-install SautiCare as the native "Senior Mode" on entry-level Android devices.

Telecom Senior Bundles

Zero-rated partnerships with **Safaricom**, **MTN**, and **Airtel**. Telcos bundle SautiCare into senior SIM packs, reducing customer care call volume and increasing mobile money transaction volume.

Diaspora Subscription

African diaspora children in the US/UK pay **\$2–\$5/month** for premium 24/7 caregiver monitoring: automated GPS alerts, battery status checks, and instant emergency dispatch for aging parents.

THE FUTURE OF VOICE IN AFRICA

No Elder in Africa Left Behind

"By replacing complex visual menus with natural, respectful voice dialogue in native African languages, SautiCare restores independence, dignity, and safety to our parents and grandparents."

Next Languages:

Hausa, Yoruba, Oromo, and
Luganda expansion.

Direct SIM Gateway:

Africa's Talking API for instant
SIM-level SMS.

Open Ecosystem:

AssemblyAI-powered
developer APIs for African
NGOs.

 **Test Live:** sauticare.onrender.com •  **GitHub:** github.com/ondinookello96/SautiCare